

Title: Methods protocol for counterfactual life table analysis of excess mortality

Authors: Jonas Schöley¹

Affiliations:

¹ Max Planck Institute for Demographic Research; Rostock, Germany

Contents

1	Summary	2
2	Data	3
2.1	Data sources	3
2.2	Data harmonization	4
2.2.1	Death count age harmonization	4
2.2.2	Exposure time calculation	5
2.2.3	Bias correction of age-specific annual death counts	5
3	Analysis	9
3.1	Counterfactual mortality forecast	9
3.2	Life table inference	9
3.3	Clustering of life expectancy deficits	11
3.4	Age decomposition analysis	12
4	Robustness and sensitivity	13
4.1	Benchmark tests	13
4.2	Alternative measures of excess mortality	15
4.3	Sensitivity of results to choice of fitting period	16
4.4	Sensitivity of results to choice of counterfactual	20
4.5	Sensitivity of results to choice of clustering procedure	24
4.6	Optimal number of clusters	27

1. Summary

We calculate life tables and associated life expectancies by sex for a selection of 34 countries for the years 2000 through 2024. For the years 2020 through 2024 we calculate counterfactual life expectancies based on the continuation of pre-pandemic mortality trends and derive associated life expectancy deficits – the difference between actual and counterfactual life expectancy – as an age-standardized and trend-adjusted measure of excess mortality.

The data basis for our calculation are midyear population and annual death counts by age, sex, and country sourced from the Short Term Mortality Fluctuations Database (HMD-STMf [1]), the Human Mortality Database (HMD, [2, 3]), the UN World Population Prospects (WPP-2024 [4]), and national statistical offices (ONS [5], CDC [6]). For a complete list of sources by country see methods appendix “Data sources”. We selected all countries represented in the Human Mortality Database for which we could find reliable death counts by age and sex for 2024 as of February 5, 2026, the date on which we froze our source data, either in the HMD’s annual data series, or in HMD-STMf. Shared characteristics of all included countries are a) that they are designated as “high-income” by the World Bank’s income classification, and b) that their data series have a high degree of reliability and comparability which is ensured by the ongoing documentation, correction, and harmonization work by the Human Mortality Database’s team.

Age-specific death counts are harmonized to a single-year age grouping with open age group 100+ using the PCLM method [7, 8]. Weekly death counts are aggregated to annual counts and midyear population estimates are converted to person-years of exposure adjusting for the effect of leap weeks. To adjust for systematic bias in registered deaths between annualized STMf data and HMD annual totals we estimate correction factors based on a statistical model (see methods appendix “Bias correction of age-specific annual death counts”).

Given the harmonized annual death counts and population exposures by age and sex we calculate period life tables using the standard piecewise-exponential model. We then fit a Poisson-Lee-Carter model [9] using the R library StMoMo [10] to the available pre-pandemic data since 2000 and forecast counterfactual age-specific mortality rates and death counts over the years 2020 through 2024. Life expectancy deficits (LED) are then derived along with P-scores [11] and the mean unfulfilled lifespan (MUL, [12]) as alternative measures of excess. We decompose the life expectancy deficits into age-specific contributions using the Arriaga method [13]. The clustering of life expectancy deficits into four groups is achieved via a hierarchical clustering algorithm implemented in the R package dtwclust [14] which we run across the country-specific trajectories (see methods appendix “Clustering of life expectancy deficits” for details).

Throughout the analysis population exposures and annual death counts by sex are aggregated as needed to derive results for both sexes combined and for the joint period 2020-2024.

We perform simulation-based uncertainty propagation throughout our analysis via a Poisson parametric bootstrap of the life table estimates and a random walk simulation of the Lee-Carter forecast. Based on this we derive a p-value for the estimated life expectancy deficit, i.e. the probability of observing or exceeding an estimated life expectancy deficit given the continuation of pre-pandemic trends (see methods appendix “Life table inference” for details).

We ensured that our findings are robust against changes in counterfactual, changes in fitting period, and changes in clustering procedure (see methods appendix “Robustness and sensitivity”). The complete analysis workflow is explained in detail in the methods appendix. The analysis script is available from <https://github.com/jschoeley/e0deficit>.

2. Data

2.1 Data sources

The inputs for our analysis are death and population counts by age, year, sex, and country. We source¹ the death counts from:

- The Human Mortality Database (HMD, [2, 3]) for annual death counts and exposures by age and sex for those countries for which a complete series is available through 2024.
- The Short Term Mortality Fluctuations Database (HMD-STMF, [1]), an input database for weekly death counts by age and sex across countries. The data are updated weekly and include the most recent information on death counts. We derive annual counts from this source in cases where the HMD does not provide data through 2024.
- The UK’s Office for National Statistics (ONS, [5]) annual death counts by age and sex for England and Wales. We use this data for years 2010 through 2019 because the age grouping in the HMD-STMF data for these years is too coarse for reliable life expectancy estimates.
- The US Centers for Disease Control and Prevention (CDC, [6]) for annual death counts by age and sex for the United States. We use this data for years 2000 through 2019 for the US, lacking any HMD-STMF data for these years.

Population data form the basis for our person-year exposure estimates and are sourced from:

- The World Population Prospects 2024 (WPP-2024, [4]) for midyear population estimates and projections by age, year, and country.
- The HMD for midyear population estimates by age, year, and country for those countries for which a complete series is available through 2024 and for the regions of the UK.

For a list of data sources by country and year see Table 1.

We selected all countries represented in the Human Mortality Database for which we could find reliable death counts by age and sex for 2024 as of February 5, 2026, the date on which we froze our source data, either in the HMD’s annual data series, or in HMD-STMF.

Countries which were excluded under these criteria are Australia (no 2024 annuals, coarse age grouping in STMF), Canada (no 2024 annuals, coarse age grouping in STMF), South Korea (no 2024 annuals, coarse age grouping in STMF), New Zealand (no 2024 annuals, coarse age grouping in STMF), Hong Kong (no 2024 annuals or STMF), Russia (no data since 2022), and Ukraine (no data since 2014).

Shared characteristics of all included countries are a) that they are designated as “high-income” by the World Bank’s income classification, and b) that their data series have a high degree of reliability and comparability which is ensured by the ongoing documentation, correction, and harmonization work by the Human Mortality Database’s team.

We excluded Scotland and Chile from the age decomposition as we found artifacts in the age patterns of mortality in both countries. For both countries we relied on the preliminary STMF data. These artifacts are localized in very old age and adolescent mortality and therefore have little influence on life expectancy estimates but some influence on the age decomposition.

¹ github.com/jschoeley/e0deficit/blob/main/src/12-download_death.R
github.com/jschoeley/e0deficit/blob/main/src/11-download_population_and_rates.R

Table 1: Data sources by country and year.

Country	Period	Death counts	Population
Austria	2000-2024	HMD-STMF	WPP-2024
Belgium	2000-2024	HMD	HMD-EST
Bulgaria	2000-2024	HMD-STMF	WPP-2024
Switzerland	2000-2024	HMD	HMD-EST
Chile	2000-2015	HMD	WPP-2024
Chile	2016-2024	HMD-STMF	WPP-2024
Czech Republic	2000-2004	-	WPP-2024
Czech Republic	2005-2024	HMD-STMF	WPP-2024
Germany	2000-2024	HMD-STMF	WPP-2024
Denmark	2000-2024	HMD	HMD-EST
Estonia	2000-2024	HMD	HMD-EST
Spain	2000-2024	HMD-STMF	WPP-2024
Finland	2000-2024	HMD	HMD-EST
France	2000-2024	HMD-STMF	WPP-2024
England and Wales	2000-2009	-	HMD-EST
England and Wales	2010-2019	ONS	HMD-EST
England and Wales	2020-2022	HMD-STMF	HMD-EST
England and Wales	2023-2024	HMD-STMF	HMD-PRJ
Northern Ireland	2000-2021	HMD	HMD-EST
Northern Ireland	2022	HMD-STMF	HMD-EST
Northern Ireland	2023-2024	HMD-STMF	HMD-PRJ
Scotland	2000-2022	HMD-STMF	HMD-EST
Scotland	2023-2024	HMD-STMF	HMD-PRJ
Greece	2000-2014	HMD	WPP-2024
Greece	2015-2024	HMD-STMF	WPP-2024
Croatia	2000	-	WPP-2024
Croatia	2001-2024	HMD-STMF	WPP-2024
Hungary	2000-2024	HMD-STMF	WPP-2024
Israel	2000-2024	HMD-STMF	WPP-2024
Iceland	2000-2024	HMD-STMF	WPP-2024
Italy	2000-2010	-	WPP-2024
Italy	2011-2024	HMD-STMF	WPP-2024
Japan	2000-2024	HMD	HMD-EST
Lithuania	2000-2024	HMD	HMD-EST
Luxembourg	2000-2024	HMD-STMF	WPP-2024
Latvia	2000-2024	HMD	HMD-EST
Netherlands	2000-2024	HMD-STMF	WPP-2024
Norway	2000-2024	HMD	HMD-EST
Poland	2000-2024	HMD-STMF	WPP-2024
Portugal	2000-2024	HMD	HMD-EST
Sweden	2000-2024	HMD	HMD-EST
Slovenia	2000-2024	HMD-STMF	WPP-2024
Slovakia	2000-2024	HMD	HMD-EST
Taiwan	2000-2024	HMD-STMF	WPP-2024
USA	2000-2019	CDC	WPP-2024
USA	2020-2024	HMD-STMF	WPP-2024

2.2 Data harmonization

2.2.1 Death count age harmonization

As the HMD-STMF input data are not harmonized in terms of age grouping we perform² a model-based ungrouping of deaths into single ages for each region-year-sex stratum using the Penalized Composite Link Method (PCLM) [7] implemented in the R package ungroup [8].

² https://github.com/jschoeley/e0deficit/blob/main/src/21-harmonize_death_with_exposures.R

The original death counts over single-year age groups $\mathbf{d}^{\text{single}} = (d_{j=1}, \dots, d_J)^\top$ have been grouped (summed) into discrete age bins $\mathbf{d}^{\text{grouped}} = (d_{i=1}, \dots, d_I)^\top$ of varying width. We assume the grouped death counts to arise from a Poisson distribution with mean μ_i :

$$d_i^{\text{grouped}} \sim \text{Poisson}(\mu_i).$$

The PCLM method seeks to recover the means of the original single-year death counts, $\boldsymbol{\gamma} = (\gamma_{j=1}, \dots, \gamma_J)^\top$. To that end, one can encode the age aggregation into a binary composition matrix $\mathbf{C}$ with dimensions $I \times J$ such that $\mathbf{d}^{\text{grouped}} = \mathbf{C}\mathbf{d}^{\text{single}}$. The grouped Poisson rates are then connected to the single age means via the composite link

$$\boldsymbol{\mu} = \mathbf{C}\boldsymbol{\gamma}.$$

By specifying that the $\boldsymbol{\gamma}$ are a smooth function of age,

$$\gamma_j = \exp(s(x(j))),$$

where $s(x(j))$ is a P-spline over the starting age x of single-year age groups $j \in 1, \dots, J$, a penalized likelihood solution can be found for the vector of mean death counts by single age. To get an integer estimate of the death count in single age x we round $\gamma_x(j)$ down to the next integer,

$$\hat{d}_x^{\text{single}} = \lfloor \gamma_x(j) \rfloor,$$

such that $\hat{d}_x$ corresponds to the mode of the Poisson distribution with rate $\gamma_x(j)$.

Because the age grouping scheme may vary within a year, the ungrouping is separately applied to each region $\times$ sex $\times$ year $\times$ age pattern combination in the input data. In a final step, the harmonized sub-annual HMD-STMF death counts are summed into annual death counts by age, $\hat{d}_{xy}$.

2.2.2 Exposure time calculation

We derive person-years of exposure by age and year, E_{xy} , from the midyear population estimates, N_{xy}^{midyear} . We choose the HMD midyear population for exposure calculation for those countries where a complete series is available through 2024 and for the regions of the UK. Otherwise we choose the World Population Prospects 2024 data, which are always complete throughout 2024. To complete HMD exposures for the UK regions we perform a stable population projection based on the most recent HMD population data and 5-year average (2015 through 2019) age-specific mortality and fertility rates. Exposures are adjusted for the length of the observation period within a year, taking into account weeks with missing data and leap weeks.

$$E_{xy} = N_{xy}^{\text{midyear}} n_y^{\text{observed-weeks}} / 52.$$

2.2.3 Bias correction of age-specific annual death counts

The purpose of our bias correction model is to correct any systematic deviations between our age-specific mortality rate estimates and the official annual estimates. While the HMD-STMF death counts allow for timely estimates of current mortality they sometimes deviate substantially from official annual estimates due to different population definitions and reporting guidelines in the weekly death reporting compared to the annual reporting. Because these systematic biases are stable in

time, they can be estimated and corrected. To that end we estimate³ a correction factor $\widehat{c}_{xy}$ which, if multiplied with the single-age annual death counts, $\widehat{d}_{xy}$, yields the corrected counts:

$$\widehat{d}_{xy}^{\text{corrected}} = \widehat{c}_{xy} \cdot \widehat{d}_{xy}.$$

We estimate the correction factor via the Poisson model

$$d_{xy}^{\text{benchmark}} \sim \text{Poisson}\left(c_{xy} \cdot \widehat{d}_{xy}\right),$$

where the benchmark death counts are taken from the HMD and

$$c_{xy} = \exp(s_y(x|y)),$$

is a smooth age effect estimated separately for each year capturing the time-varying deviations between unadjusted and benchmark death counts. The correction factor from the last year with available benchmark data will be used in subsequent years.

Figures 1 and 2 show the estimated correction factors and the bias-corrected age-specific death counts by year for the example of Austrian females. The model is fit separately to each region and sex.

³ https://github.com/jschoeley/e0deficit/blob/main/src/31-rate_bias_correction.R

Figure 1: Demonstration of estimated (red) bias correction factors c_{xy} vs. raw correction factors (black) for Austrian females.

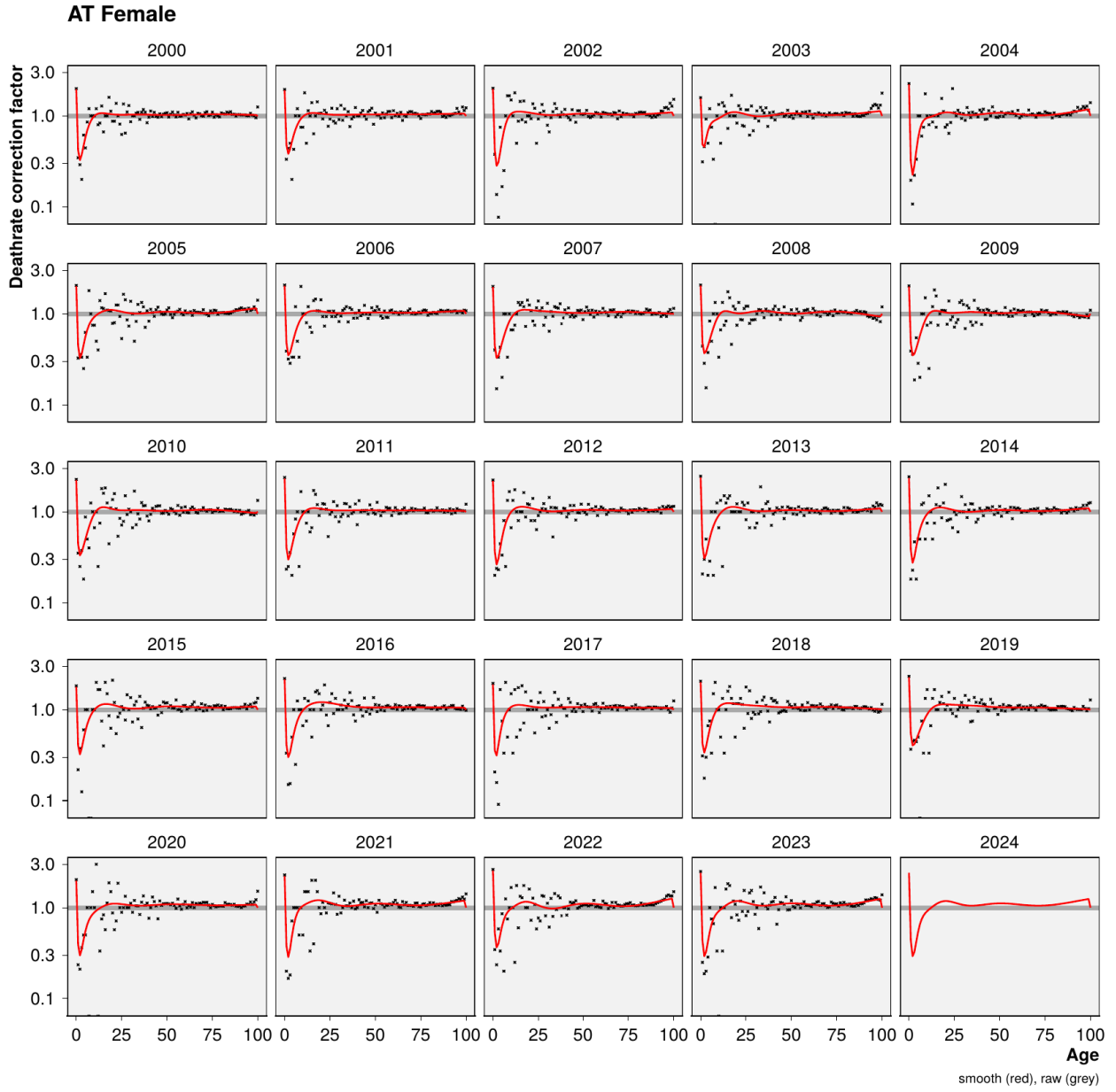
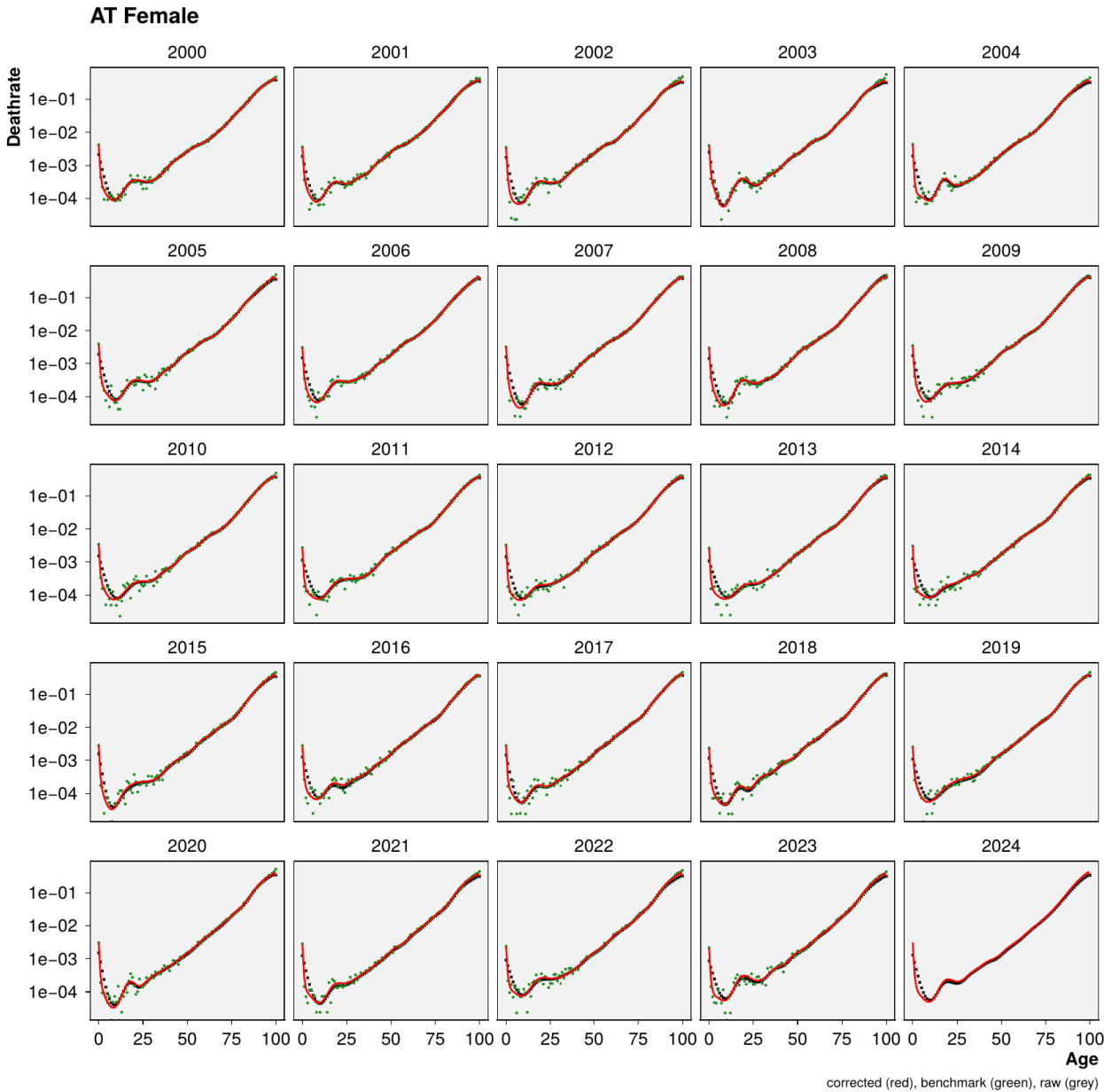


Figure 2: Demonstration of bias-corrected ungrouped age-specific death rates (red) vs. unadjusted death rates (black) and HMD benchmark death rates (green) for Austrian females.



3. Analysis

3.1 Counterfactual mortality forecast

In order to calculate the life expectancy under continuation of pre-pandemic trends we forecast⁴ age-specific mortality rates over the years 2020-2024 using a Poisson-Lee-Carter model given the data available up to January 2020.

We fit a Poisson-Lee-Carter model [9] where death counts at single age x and single year y are distributed

$$d_{xy} \sim \text{Poisson}(\mu_{xy} E_{xy}),$$

with death rates

$$\mu_{xy} = \exp(\alpha_x + \beta_x \kappa_y),$$

and identifiability constraints $\sum_y \kappa_y = 0$ and $\sum_x \beta_x = 1$.

We forecast μ_{xy} past the fitting period via the usual random walk with drift over κ_y ,

$$\kappa_{y+1} = \kappa_y + \hat{b} + \text{Normal}(0, \hat{\sigma}).$$

We estimate the variance and mean difference of the estimated year-on-year κ_y changes and simulate a corresponding random walk with drift over the years 2020-2024. The resulting forecasts of κ are then used to derive a corresponding surface of age-period mortality rates $\mu_{xy}^{\text{expected}} = \exp(\alpha_x + \beta_x \kappa_y)$ for forecast horizon $y \in \{2020, 2021, \dots, 2024\}$. We fit the model using the R library StMoMo [10]. Forecast uncertainty is propagated throughout the rest of the analysis via 250 samples of the forecasted $\mu_{xy}^{\text{expected}}$.

3.2 Life table inference

We perform a parametric bootstrap of our life table estimates⁵ based on 250 Poisson replicates of actual and expected age-specific death counts for any population under consideration (for brevity sex, country, and year indices are omitted in this section).

$$d_x^{\text{actual}} \sim \text{Poisson}(\hat{d}_x^{\text{corrected}})$$

$$d_x^{\text{expected}} \sim \text{Poisson}(\mu^{\text{expected}} \cdot E_x)$$

At this point we aggregate the simulated death counts along with the person-years of exposure into larger populations of interest, e.g. male and female are summed to total population and annual data are summed to multi-year aggregates. All of the following analysis steps are also performed for the newly created aggregated populations.

For any given population and for each sample draw we calculate the actual and expected central death rates by age as

⁴ https://github.com/jschoeley/e0deficit/blob/main/src/40-counterfactual_projection.R

⁵ https://github.com/jschoeley/e0deficit/blob/main/src/50-estimation_and_inference.R

$$m_x^{\text{actual}} = d_x^{\text{actual}} / E_x,$$

and

$$m_x^{\text{expected}} = d_x^{\text{expected}} / E_x.$$

These rates are then transformed into life tables yielding actual and expected life expectancies, e_x^{actual} and e_x^{expected} , for each population, year, and simulation draw. We write $n_x = 1$ for $x \neq \omega$ and $n_\omega = \infty$ for the width of the age group starting at x with $\omega = 100$ being the open age group closing the life table. Under the piecewise constant assumption we calculate the life tables as

$$\begin{aligned} \Pr(X \geq x + n | X > x) = {}_n p_x &= \begin{cases} 0 & \text{if } x = \omega \\ \exp(-{}_n m_x n_x) & \text{otherwise} \end{cases} \\ \Pr(X > x) = \ell_x &= \begin{cases} 1 & \text{if } x = 0 \\ \prod_{i=0}^{x-1} {}_n p_{x=i} & \text{otherwise} \end{cases} \\ \Pr(x \leq X < x + n) = {}_n d_x &= \begin{cases} \ell_\omega & \text{if } x = \omega \\ \ell_{x+n} - \ell_x & \text{otherwise} \end{cases} \\ \int_x^{x+n} \ell_a \, da = {}_n L_x &= \begin{cases} 1 / {}_n m_x & \text{if } x = \omega \\ {}_n d_x / {}_n m_x & \text{otherwise} \end{cases} \\ \int_x^\infty \ell_a \, da = T_x &= \sum_{i=x}^{\omega} {}_n L_{x=i} \\ \mathbb{E}(X | X > x) = e_x &= T_x / \ell_x. \end{aligned}$$

Now the *life expectancy deficit* can be calculated as the difference between actual and counterfactual life expectancy, the baseline life expectancy under the continuation of pre-pandemic trends,

$$e_0^{\text{deficit}} = e_0^{\text{actual}} - e_0^{\text{expected}}.$$

Note that we define the deficit such that it is negative if actual life expectancy falls short of the expectation, i.e. if there are excess deaths, and positive if there is a life expectancy surplus, i.e. if the expectation is outperformed.

Life expectancy deficits do not necessarily indicate a structural deviation from expected mortality trends as both the expected and the observed life tables are subject to uncertainty. Uncertainty in the expectation is mainly due to forecasting error – we do not know exactly how age-specific mortality rates would have developed in the absence of the pandemic. This uncertainty is captured by the random walk of the forecasting model described earlier. Additionally, aleatoric uncertainty in the death counts induces uncertainty in the estimated central death rates which is reflected via our Poisson resampling scheme.

Due to the stated sources of randomness one may observe life expectancy deficits even under the continuation of pre-pandemic trends. Thus we perform inference about the estimated e_0^{deficit} by calculating a p-value stating the probability of estimating a life expectancy deficit as large as the realized deficit under the continuation of pre-pandemic trends. Life expectancy deficits under the null hypothesis of pre-pandemic trends are defined as

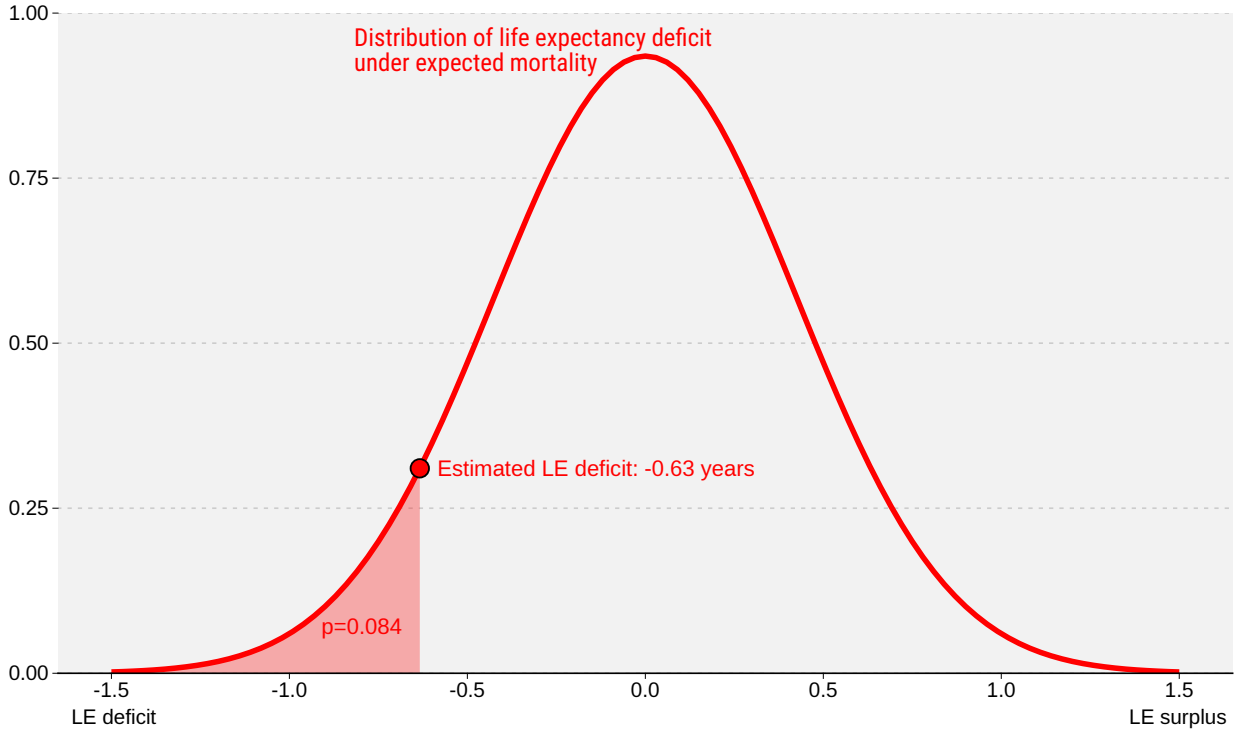
$$T = e_0^{\text{expected}} - \mathbb{E}(e_0^{\text{expected}}),$$

and we derive the sampling distribution of T from our sample draws of e_0^{expected} . We can then calculate the probability of our calculated e_0^{deficit} given pre-pandemic trends as

$$\Pr(e_0^{\text{deficit}} \leq T^*).$$

Figure 3 demonstrates the procedure.

Figure 3: Demonstration of p-value calculation for estimated life expectancy deficit.



3.3 Clustering of life expectancy deficits

We calculate time series of life expectancy deficits, $e_{0,y}^{\text{deficit}}$, for years 2020 through 2024 by region. These series are then clustered⁶ into four groups: A) Early deficit (peak deficit in 2020 or 2021 without pronounced worsening in 2021), B) Second wave peak deficit (pronounced peak deficit in 2021), C) Late deficit (worsening of deficits starting 2022), and D) Prolonged deficit (deficits since 2020 without pronounced peak or trend).

The group assignment for each country's deficit trajectory is formalized by employing hierarchical clustering with a pre-specified cluster number of four. We cluster on deficit series for both sexes combined. Prior to clustering we derive three time series features for each region c : the range of the deficits, $r_c = \max_y(e_{0,y,c}^{\text{deficit}}) - \min_y(e_{0,y,c}^{\text{deficit}})$, the centered and range-normalized trajectory of deficits, $s_{y,c} = \left(e_{0,y,c}^{\text{deficit}} - \text{mean}_y(e_{0,y,c}^{\text{deficit}}) \right) / r_c$, and the first differences of $s_{y,c}$ over time. We then run the hierarchical clustering algorithm implemented in the R package `dtwclust` [14] with Euclidean

⁶ https://github.com/jschoeley/e0deficit/blob/main/src/51-assign_clusters.R

distance and Ward’s minimum variance method on the combined vector of features across regions to determine each region’s cluster. The resulting clustering captures the phenomenology of the four qualitative groups described above.

3.4 Age decomposition analysis

We perform the Arriaga decomposition [13] to attribute life expectancy deficits to age-specific excess mortality⁷. As the Arriaga decomposition is analytic and can be expressed as a series of simple arithmetic operations it lends itself well to simulation-based inference where computation speed is vital. Based on the previously estimated life table quantities we calculate

$$\begin{aligned} {}_n\text{DE}_x &= \left({}_nL_x^{\text{actual}} / \ell_x^{\text{actual}} - {}_nL_x^{\text{expected}} / \ell_x^{\text{expected}} \right) \cdot \ell_x^{\text{actual}} \\ {}_n\text{IE}_x &= \left(\ell_x^{\text{expected}} / \ell_x^{\text{actual}} - \ell_{x+n}^{\text{expected}} / \ell_{x+n}^{\text{actual}} \right) \cdot T_{x+n}, \end{aligned}$$

which yields the contribution of excess mortality at age x to the overall life expectancy deficit:

$${}_n\Delta_x^{e_0} = {}_n\text{DE}_x + {}_n\text{IE}_x.$$

⁷ https://github.com/jschoeley/e0deficit/blob/main/src/50-estimation_and_inference.R

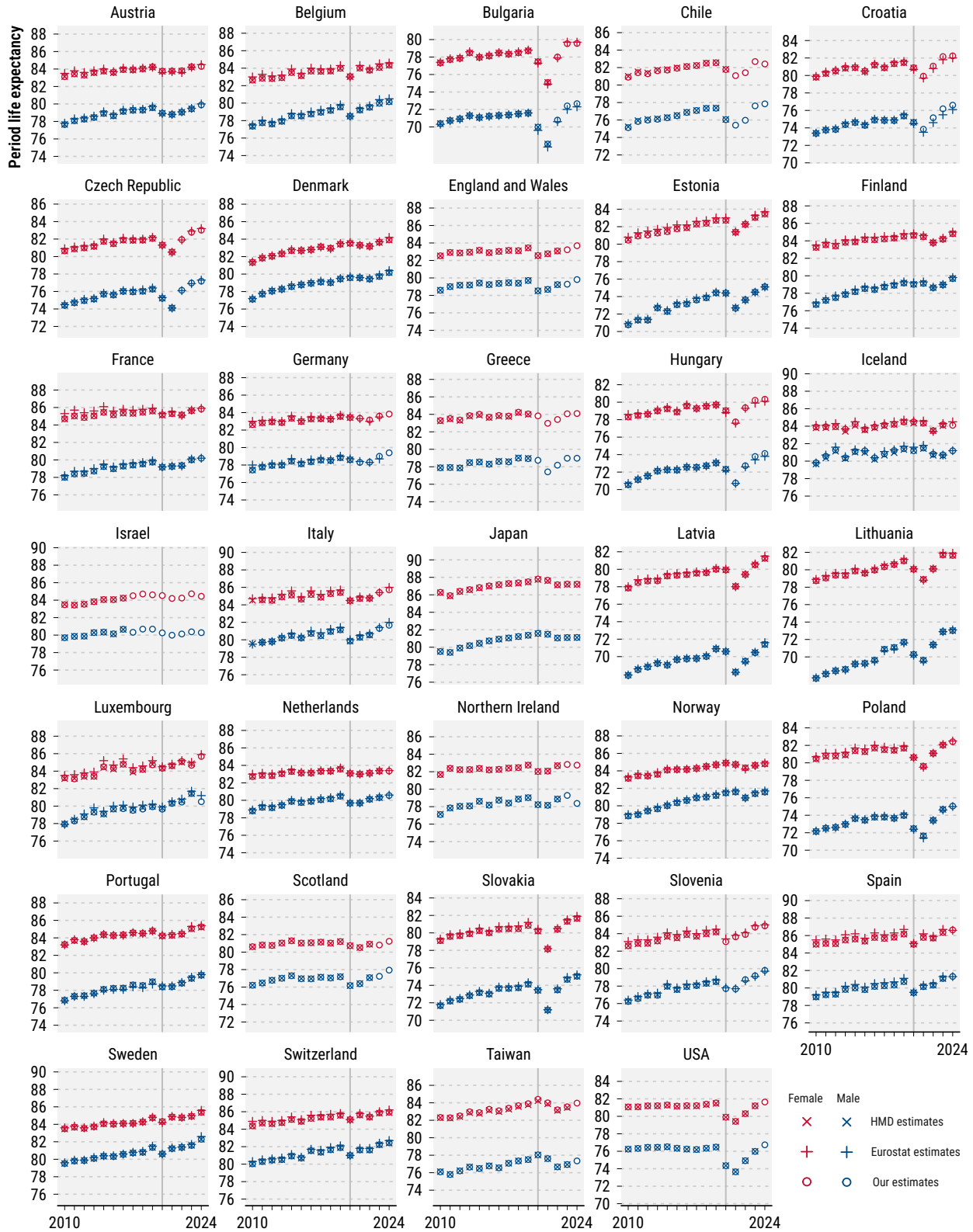
4. Robustness and sensitivity

4.1 Benchmark tests

We compare⁸ our life expectancy estimates to available estimates by Eurostat and the Human Mortality Database. As intended, our estimates are closely aligned with the HMD reported life expectancies. In some regions, Spain for example, small discrepancies exist between HMD's and Eurostat's estimates. However, the overall shape of the life expectancy trajectories is qualitatively the same under either set of estimates.

⁸ https://github.com/jschoeley/e0deficit/blob/main/src/91-benchmark_check_e0.R

Figure 4: Comparison of our life expectancy estimates with past estimates by Eurostat and the Human Mortality Database.



4.2 Alternative measures of excess mortality

In addition to the life expectancy deficit we calculate⁹ alternative measures of excess mortality as a sensitivity check.

Excess deaths are the difference between actual and expected deaths,

$$\text{Excess}_x = d_x^{\text{actual}} - d_x^{\text{expected}}$$

and can be turned into the P-score[11] by aggregating over age and expressing them as a ratio to total expected deaths,

$$\text{Pscore} = \sum_x \frac{\text{Excess}_x}{d_x^{\text{expected}}}.$$

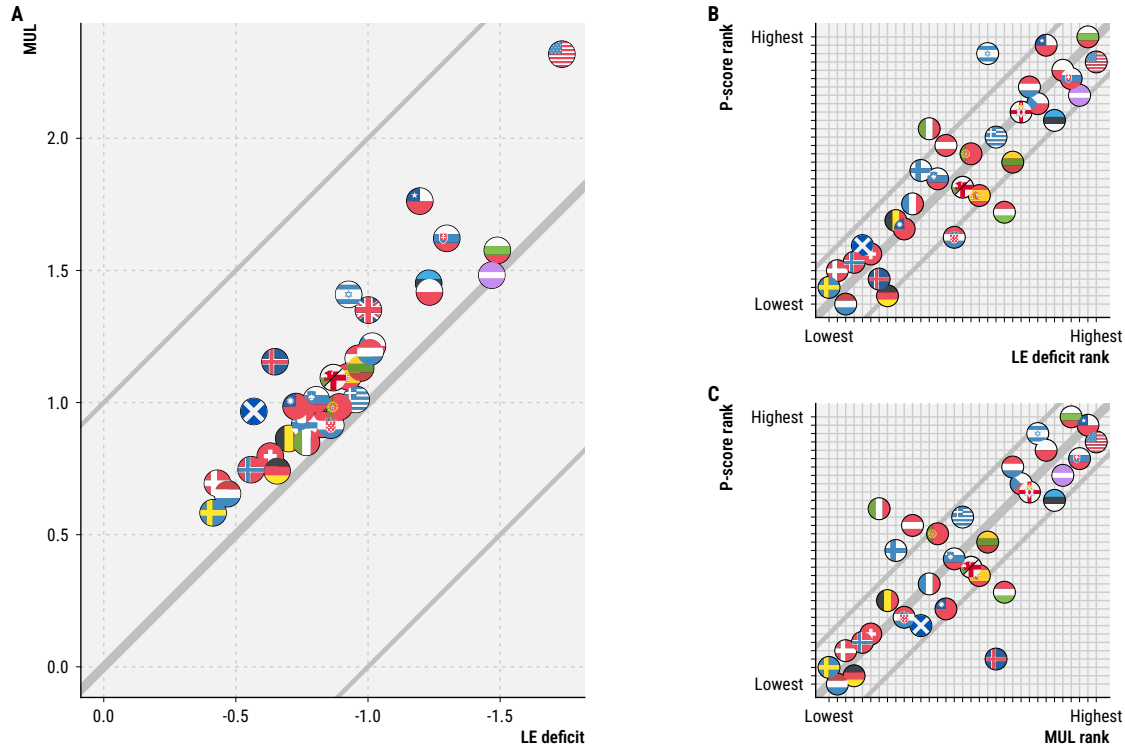
The mean unfulfilled lifespan (MUL, [12]), like the life expectancy deficit, is based on counterfactual life expectancies. MUL is defined as the difference between the average age at death during some period and the average expected age at death under counterfactual mortality conditions.

$$\text{MUL} = \frac{1}{\sum_x d_x^{\text{actual}}} \sum_x \text{Excess}_x \cdot e_x^{\text{expected}}.$$

Figure 5 shows a comparison of the three measures with each other across countries.

⁹ https://github.com/jschoeley/e0deficit/blob/main/src/92-sensitivity_check_excess_measures.R

Figure 5: Cross-country comparison of alternative excess mortality measures for the period 2020-2024. A) Life expectancy deficit vs. mean unfulfilled lifespan, B) Country ranking of life expectancy deficit vs. P-score, C) Country ranking of mean unfulfilled lifespan vs. P-score.



4.3 Sensitivity of results to choice of fitting period

As a robustness check we recalculated key results in our paper under a 10-year fitting period for the counterfactual life expectancy. Key results remain robust under either 10 or 20 years of fitting while the existence of significant life expectancy deficits in 2024 changes for some countries.

The cluster assignment under the hierarchical clustering algorithm is perfectly robust against changes in the fitting period with identical clusters identified under either analysis. Visually, the salient features of each cluster remain robust. A 10-year fitting period yields slightly lower life expectancy deficits 2020-2024 compared to the longer fit (see Figures 6 and 7).

Figure 6: Deficit typology under 20-year Lee-Carter fitting period.

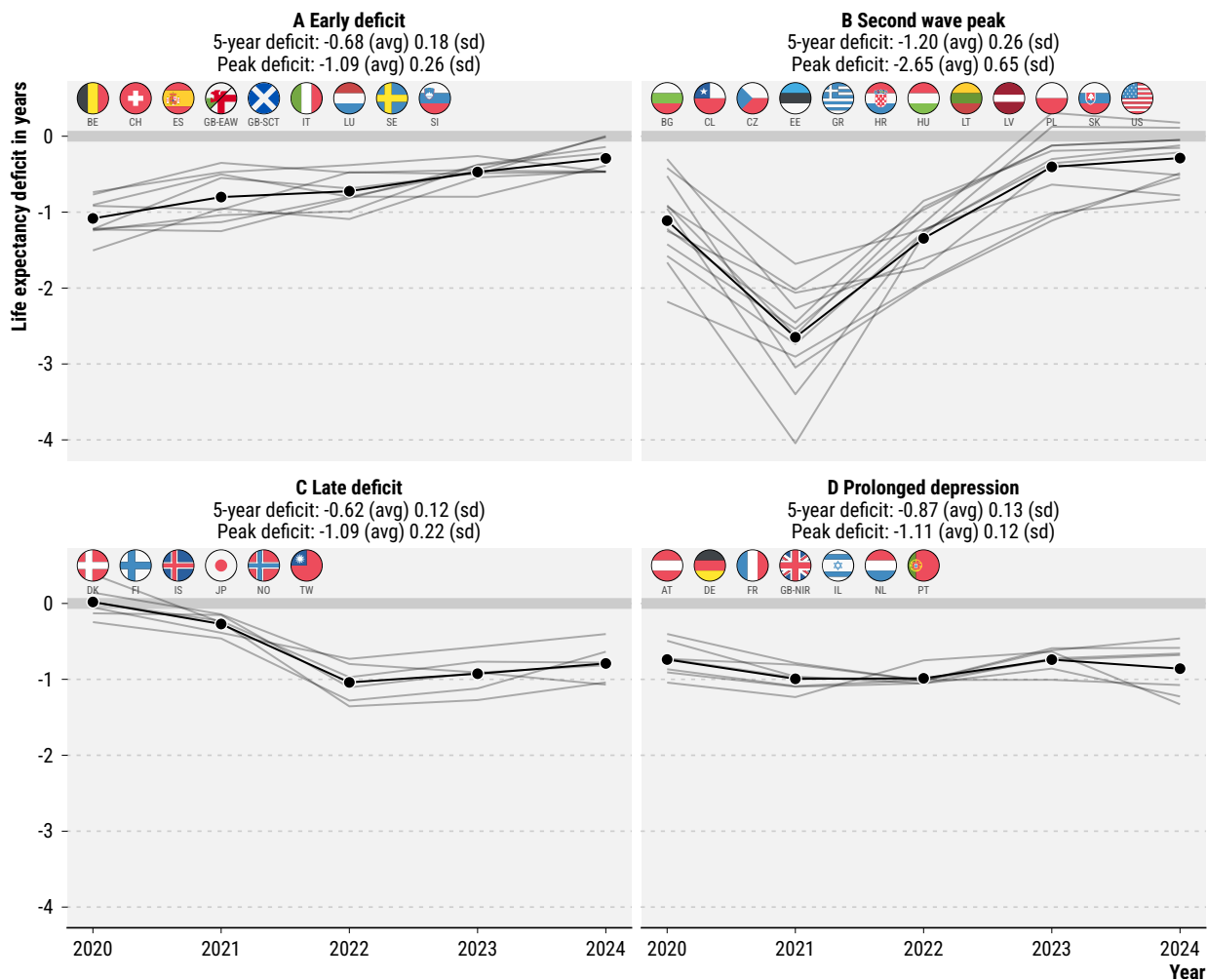
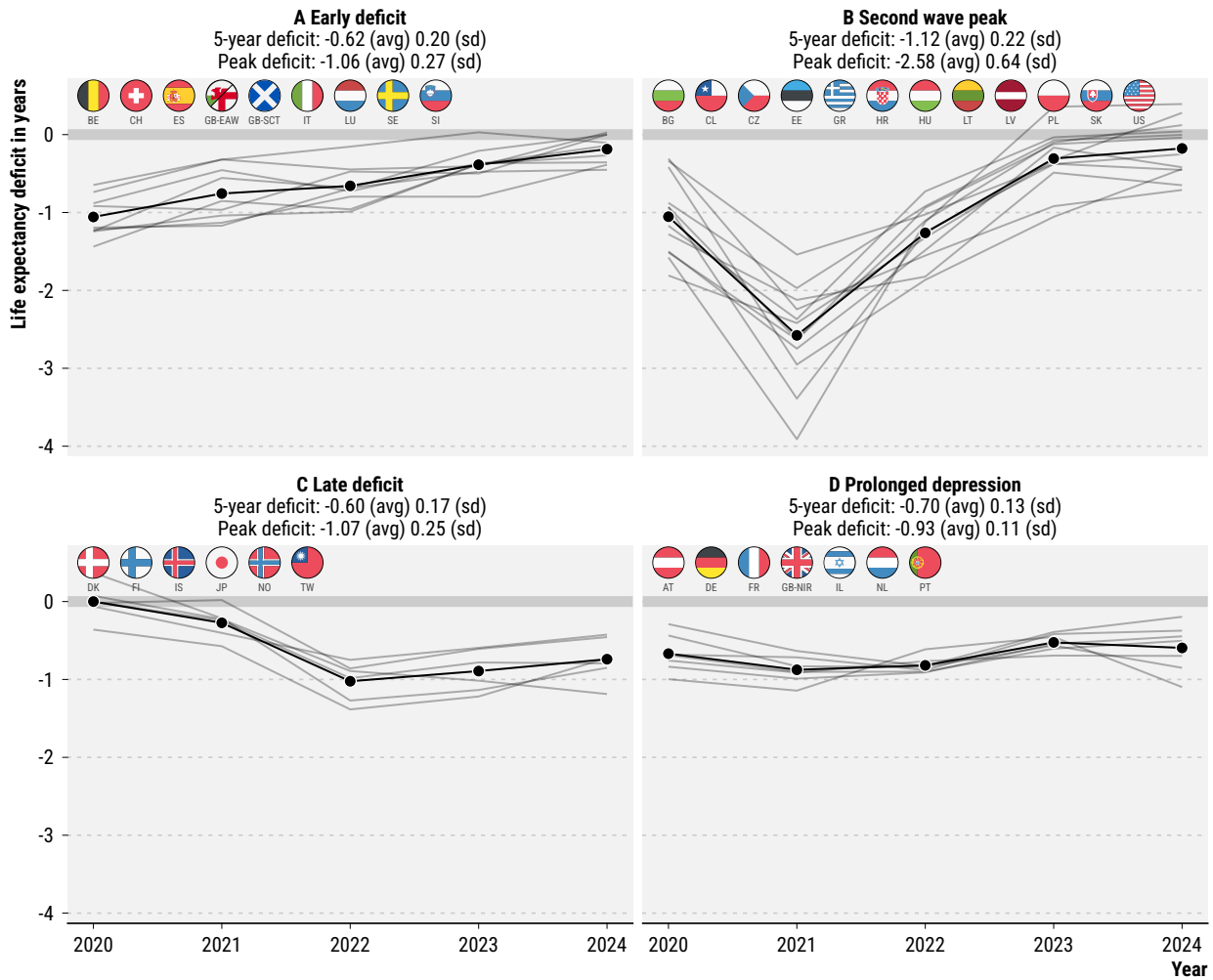


Figure 7: Deficit typology under 10-year Lee-Carter fitting period.



Under a 20-year fitting period we estimated 12 countries (AT, CL, DK, FI, GB-NIR, GR, IL, JP, NL, NO, TW, US) with a significant ($p < 0.05$) life expectancy deficit in 2024 compared to 9 (CL, DK, FI, GB-NIR, IL, JP, NL, NO, TW) under a 10-year fit.

The US is uniquely sensitive to the choice of fitting period due to a period of life expectancy stagnation pre-2020. A 10-year fitting window continues the flat trajectory, leading to lower life expectancy deficits, whereas a 20-year fitting window extrapolates a longer-term trend of slight life expectancy increase. As a result the finding that the US had the highest deficit overall is not robust (see Figures 8 and 9).

Figure 8: Annual life expectancy deficits under 20-year Lee-Carter fitting period.

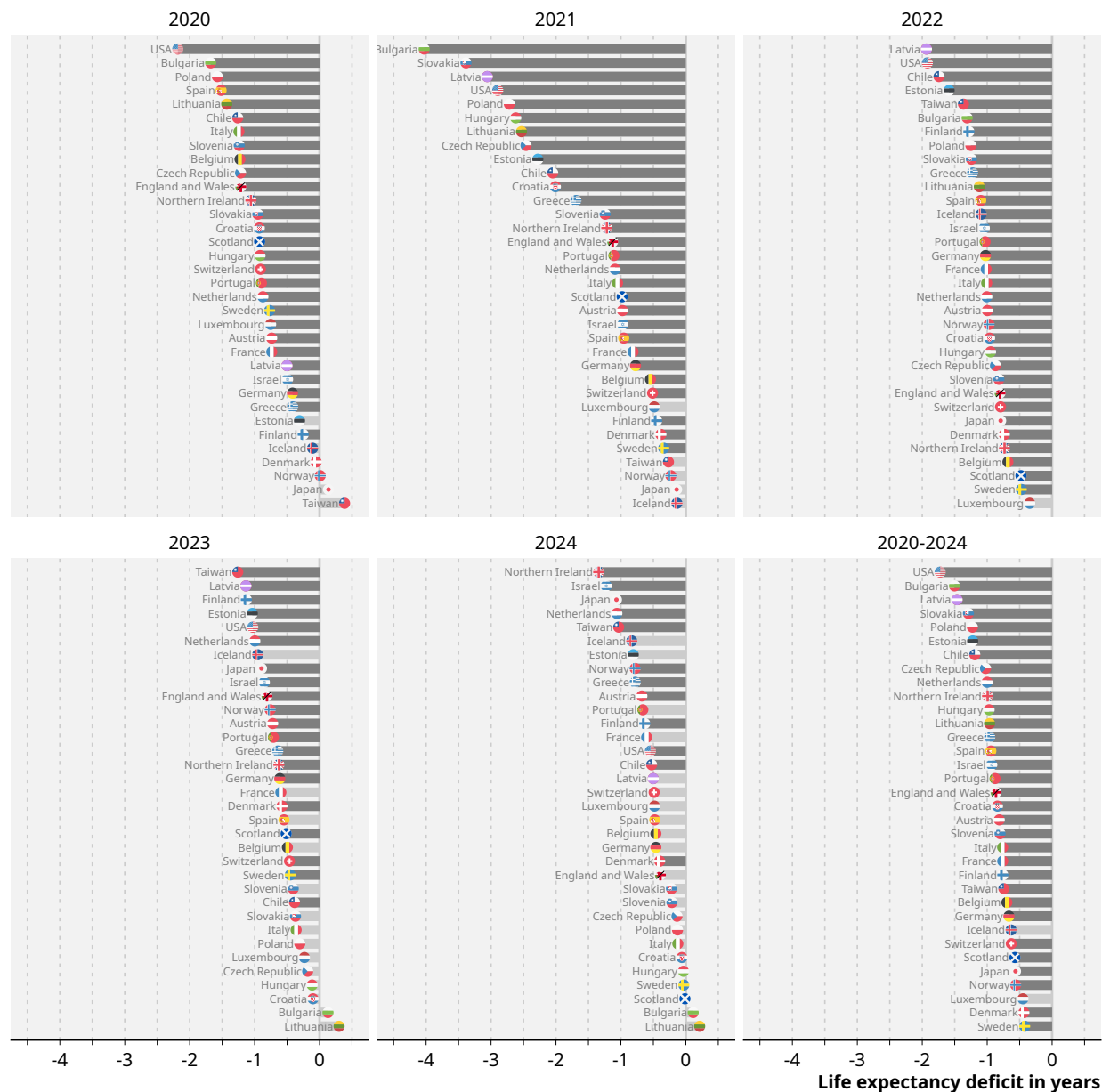
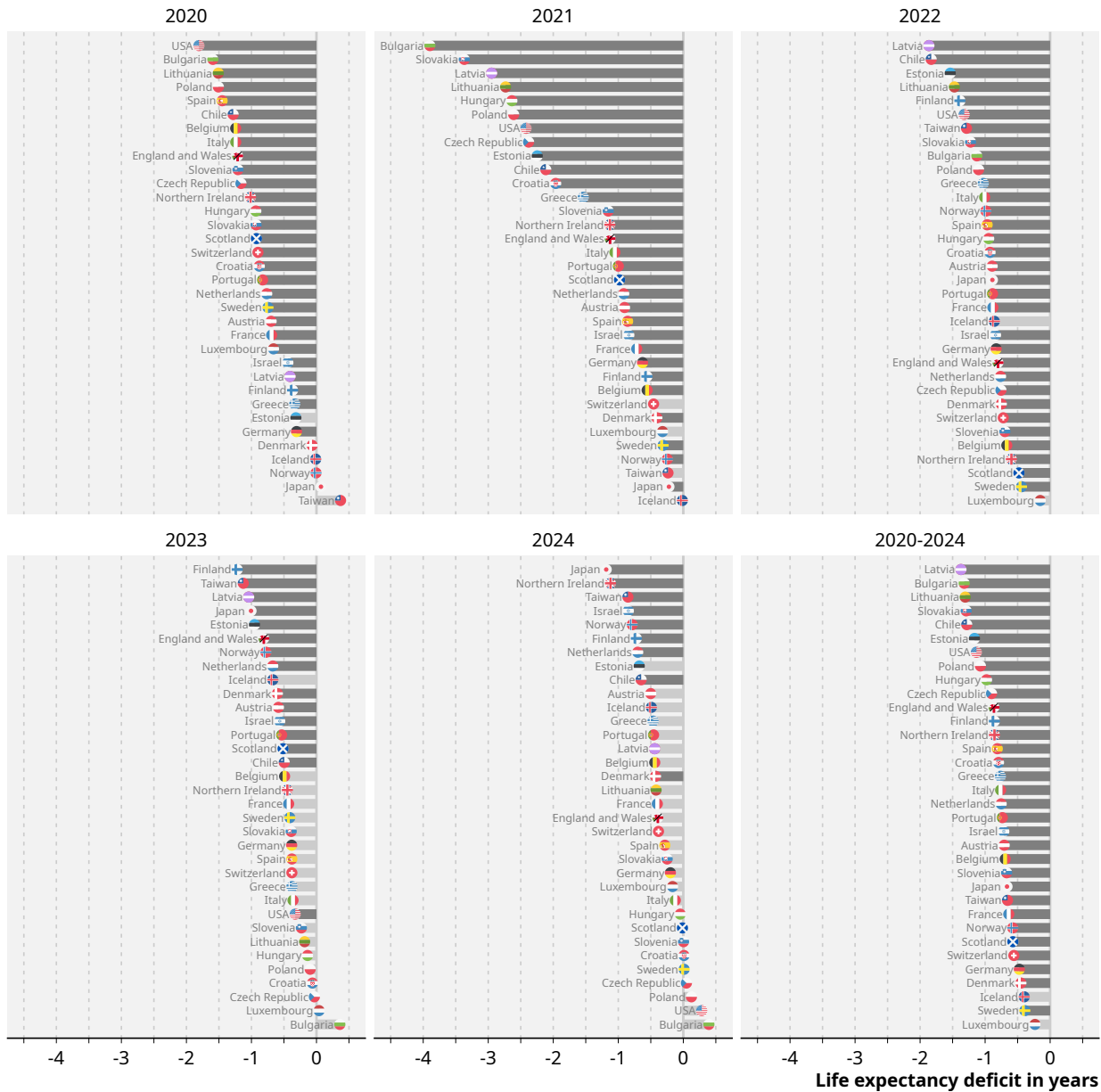


Figure 9: Annual life expectancy deficits under 10-year Lee-Carter fitting period.



4.4 Sensitivity of results to choice of counterfactual

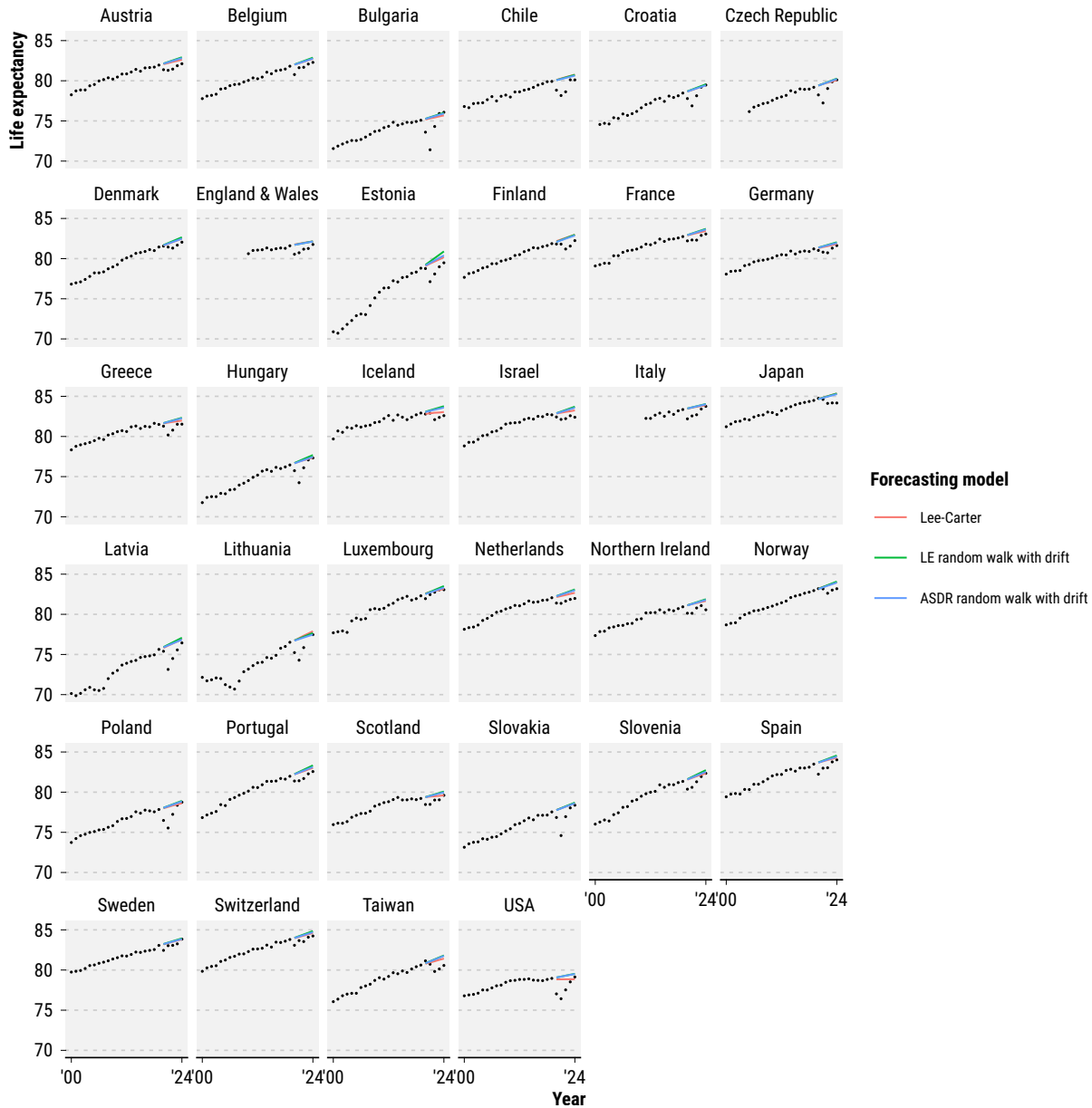
The choice of model for producing the counterfactual life expectancy under continuation of pre-pandemic trends influences life expectancy deficit estimates. Here we compare¹⁰ our results under three different counterfactuals a) the Lee-Carter forecasts used in the main text, b) a direct extrapolation of 2019 life expectancy under continuation of the mean annual changes observed over 2000-2019 (LE random walk with drift), and c) a log-linear extrapolation of age-specific death rates based on mean annual changes 2000-2019 (ASDR random walk with drift).

Figure 10 displays the counterfactual life expectancy forecasts under different models. For the majority of locations the results are coherent. In countries where forecasts diverge noticeably, such as Estonia, Iceland, Israel, the Netherlands, Scotland, and the US, the Lee-Carter model is always

¹⁰ https://github.com/jschoeley/e0deficit/blob/main/src/95-sensitivity_check_model_choice.R

the least optimistic one, i.e. it produced lower counterfactual life expectancies compared with the alternative approaches. This tendency of the Lee-Carter model to be conservative in its forecasts makes our findings of life expectancy deficits conservative as well. The deficits we find tend to be even larger under alternative models.

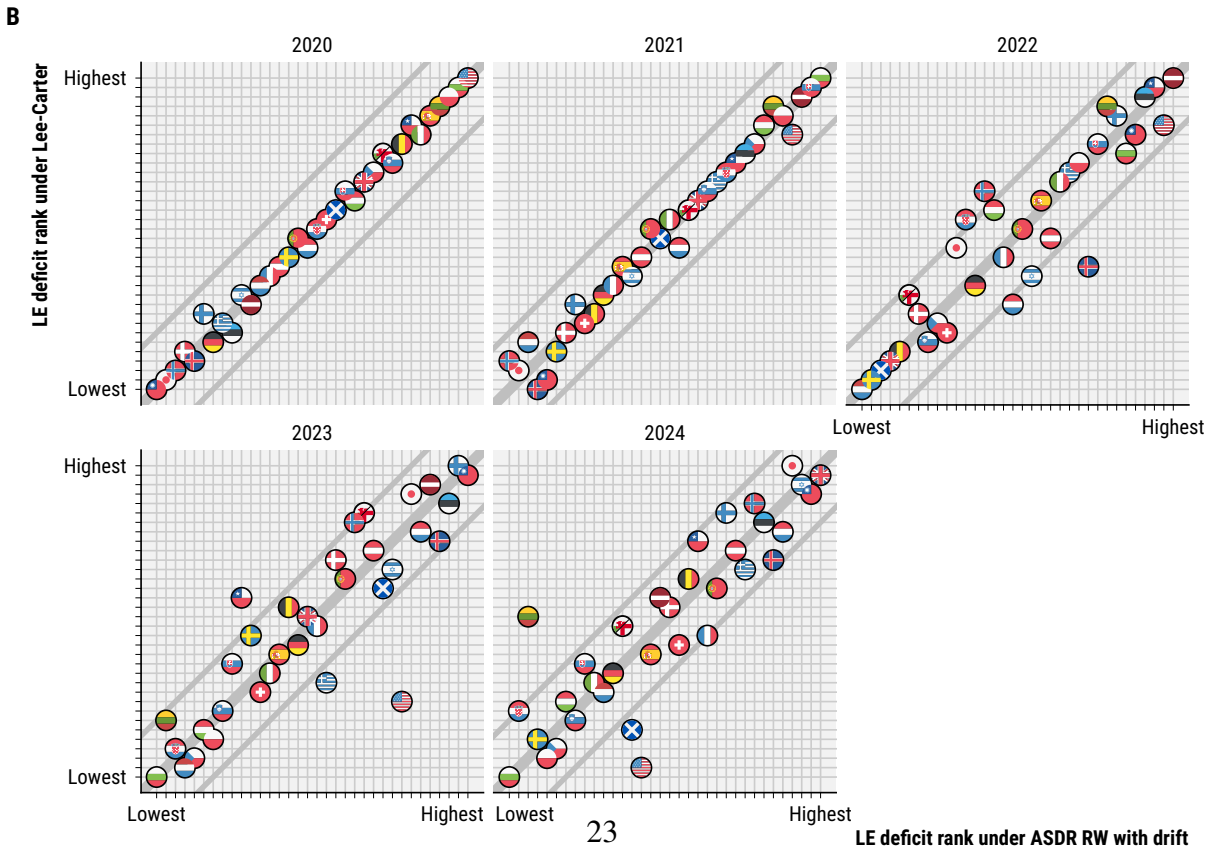
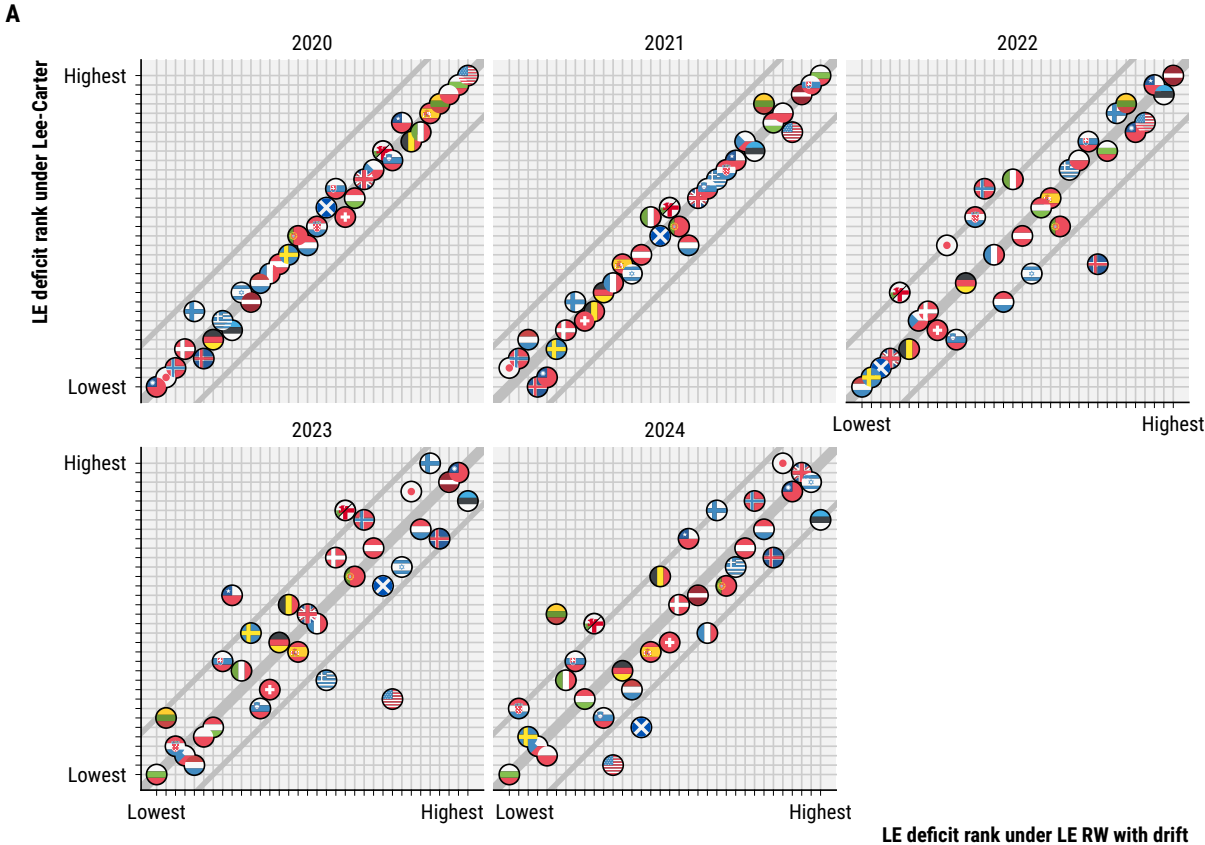
Figure 10: Comparison of life expectancy forecasts 2020-2024 under three different models.



The country ranking in annual life expectancy deficits is somewhat sensitive to the choice of counterfactual model and becomes more sensitive over time as shown in Figure 11. However, almost all countries are within ± 5 ranks of their Lee-Carter rank when comparing against the two alternative counterfactuals. Notable exceptions are the US in 2023 and 2024 and Iceland in 2022. These changes in rank under different counterfactual life expectancy trajectories are, however, consistent with the prediction intervals given by the Lee-Carter method, which also derive from a set of simulated plausible counterfactuals. The prediction intervals around life expectancy deficits are wide

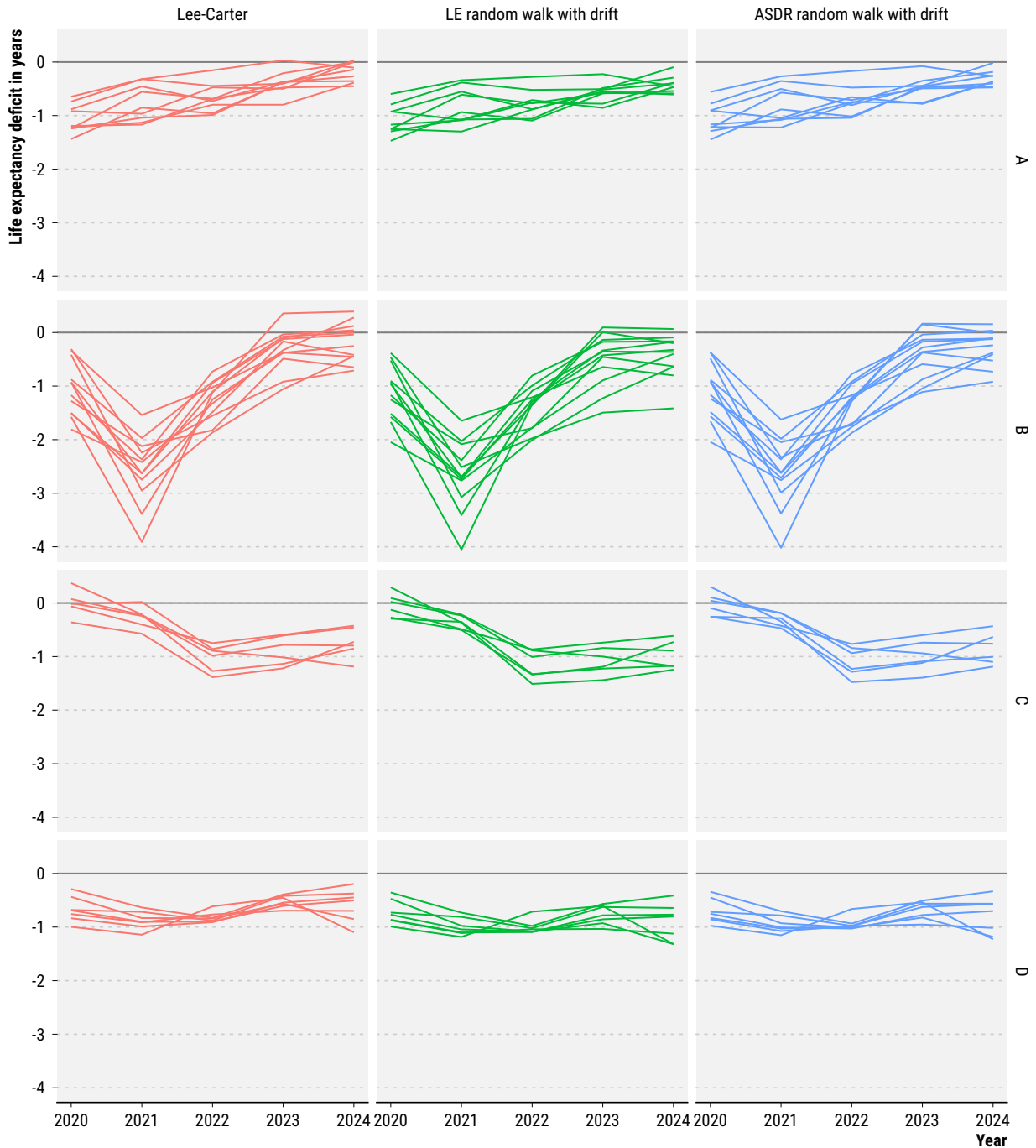
enough to reliably identify the top, middle, and bottom groups in cross-country comparisons of life expectancy deficits, but do not allow for the robust identification of exact ranks.

Figure 11: Country ranking of annual life expectancy deficits under different counterfactual models. Comparison of the Lee-Carter counterfactual with A) a direct extrapolation of life expectancy under continuation of average annual rate of increase over the time period 2000-2019, and B) a log-linear extrapolation of age-specific death rates based on average changes over the time period 2000-2019.



In Figure 12 we apply the same clustering as in the main text to life expectancy deficits under each counterfactual. Clearly the main cluster features reproduce almost perfectly across the different counterfactuals.

Figure 12: Annual life expectancy deficits and clusters under different counterfactual models.



4.5 Sensitivity of results to choice of clustering procedure

To establish robustness, we compare the group assignment under three different clustering procedures¹¹.

¹¹ https://github.com/jschoeley/e0deficit/blob/main/src/51-assign_clusters.R

Hierarchical clustering

The algorithm is the same as described in section 3.3 and as used in the main text.

Probabilistic decision rule

A similar cluster assignment as under the hierarchical algorithm can be achieved with a much simpler method of decision rules over our posterior simulation draws. First we test if a country's life expectancy deficit series features a significant ($\alpha = 0.1$) peak or slope. If no, we assign the group D, otherwise we check if the probability for 2021 being the peak deficit year is $\geq 90\%$. If yes, we assign the group B, otherwise we check if the mean peak deficit was later than 2021. If yes we assign group C, otherwise group A. This method achieves an overlap of 85.3% in cluster assignment with the hierarchical clustering method.

Non-probabilistic decision rule

An even simpler, entirely non-probabilistic, algorithm for cluster assignment can be defined by labeling as group D every country where the deficit peak is less than 0.2 years above the deficit in neighboring years and the range is less than 0.75 years. For all remaining countries the year of peak deficit determines group A (2020), group B (2021), or group C (later than 2021). This simplistic method assigns the same label as the hierarchical clustering method in 82.4% of regions.

Table 2 shows the cluster assignments per country under the three different clustering procedures. The results in the paper are reported under the hierarchical clustering.

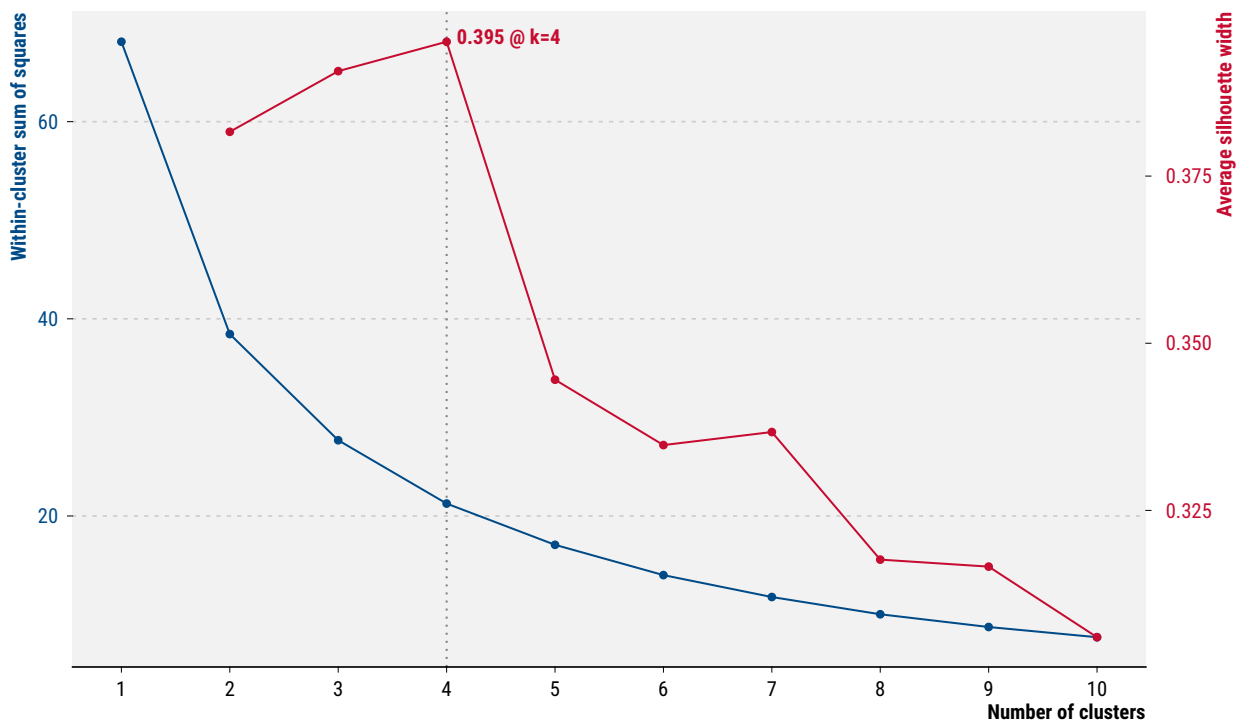
Table 2: Regional cluster assignments under different clustering procedures.

Region	Hierarchical	Probabilistic	Non-probabilistic
AT	D	D	D
BE	A	A	A
BG	B	B	B
CH	A	A	A
CL	B	B	B
CZ	B	B	B
DE	D	C	C
DK	C	C	C
EE	B	B	B
ES	A	A	A
FI	C	C	C
FR	D	C	C
GB-EAW	A	A	A
GB-NIR	D	C	C
GB-SCT	A	A	B
GR	B	B	B
HR	B	B	B
HU	B	B	B
IL	D	C	C
IS	C	C	C
IT	A	A	A
JP	C	C	C
LT	B	B	B
LU	A	D	A
LV	B	B	B
NL	D	D	D
NO	C	C	C
PL	B	B	B
PT	D	D	D
SE	A	A	A
SI	A	A	B
SK	B	B	B
TW	C	C	C
US	B	B	B

4.6 Optimal number of clusters

Figure 13 contrasts the within cluster sum of squares with the average silhouette width across different cluster counts. The model evaluated is the hierarchical clustering as used in the main text of the paper. Naturally, the within cluster variability decreases with more clusters. However, this comes at the expense of clusters becoming more similar to each other. The silhouette width [15] measures the balance between within cluster coherence and between cluster separation. An optimum is reached with four clusters.

Figure 13: Evaluation of cluster coherence and separation of the hierarchical model over different cluster counts.



References

- [1] D. A. Jdanov, A. A. Galarza, V. M. Shkolnikov, D. Jasilionis, L. Németh, D. A. Leon, C. Boe, and M. Barbieri, “The short-term mortality fluctuation data series, monitoring mortality shocks across time and space,” *Scientific Data*, vol. 8, p. 235, Sept. 2021.
- [2] Max Planck Institute for Demographic Research, University of California, and French Institute for Demographic Studies, “Human mortality database,” 2025.
- [3] M. Barbieri, J. R. Wilmoth, V. M. Shkolnikov, D. Gleij, D. Jasilionis, D. Jdanov, C. Boe, T. Riffe, P. Grigoriev, and C. Winant, “Data resource profile: The human mortality database (HMD),” *International Journal of Epidemiology*, vol. 44, pp. 1549–1556, June 2015.
- [4] United Nations Department for Economic and Social Affairs, *World Population Prospects 2024: Summary of Results*. United Nations, 2025.
- [5] Office for National Statistics, “Deaths registered in England and Wales 2019 edition,” 2025.
- [6] Centers for Disease Control and Prevention, “Underlying Cause of Death 1999-2019 CDC WONDER Online Database,” 2025.
- [7] S. Rizzi, J. Gampe, and P. H. C. Eilers, “Efficient estimation of smooth distributions from coarsely grouped data,” *American Journal of Epidemiology*, vol. 182, pp. 138–147, July 2015.
- [8] M. D. Pascariu, M. J. Daňko, J. Schöley, and S. Rizzi, “Ungroup: An R package for efficient estimation of smooth distributions from coarsely binned data,” *The Journal of Open Source Software*, 2018.
- [9] R. D. Lee and L. R. Carter, “Modeling and forecasting U.S. mortality,” *Journal of the American Statistical Association*, vol. 87, pp. 659–671, Sept. 1992.
- [10] A. Villegas, V. K. Kaishev, and P. Millossovich, “StMoMo: An R Package for Stochastic Mortality Modelling,” *SSRN Electronic Journal*, 2015.
- [11] N. Ullrich-Kniffka and J. Schöley, “Population age structure dependency of the excess mortality P-score,” *Population Health Metrics*, vol. 22, p. 25, Sept. 2024.
- [12] P. Heuveline, “The Mean Unfulfilled Lifespan (MUL): A new indicator of the impact of mortality shocks on the individual lifespan, with application to mortality reversals induced by COVID-19,” *PLOS ONE*, vol. 16, p. e0254925, July 2021.
- [13] E. E. Arriaga, “Measuring and explaining the change in life expectancies,” *Demography*, vol. 21, pp. 83–96, Feb. 1984.
- [14] Alexis Sarda-Espinosa, “Dtwclust: Time Series Clustering Along with Optimizations for the Dynamic Time Warping Distance,” 2024.
- [15] P. J. Rousseeuw, “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis,” *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, Nov. 1987.